Calibrated Discretization and the Hodge Conjecture: The Proof Spine of a Four-Paper Series

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Abstract

Let X be a smooth complex projective manifold of complex dimension n. Assuming the interface theorems established in three companion manuscripts on local holomorphic manufacturing, coherent packet assembly, and exact-class gluing/rounding [1, 2, 3], we derive the final Hodge conclusion for rational (p,p)-classes. The argument has three stages: (1) a signed decomposition reduces any rational Hodge class to the cone-positive case; (2) for cone-positive classes, the imported discretization machine produces integral cycles with vanishing calibration defect in a fixed homology class; (3) Federer–Fleming compactness yields a calibrated holomorphic limit, which is algebraic by Harvey–Lawson/King and Chow/GAGA.

This paper therefore contains only the proof spine of the series: the closure chain from the interface theorems to the final Hodge conclusion. The local holomorphic manufacturing, coherent discretization, and exact-class gluing machinery are not reproved here; they are used only through the explicit interface statements of Section 2.

1 Introduction

We formulate the Hodge problem for a fixed rational (p,p) class on a smooth complex projective manifold and outline the closure argument proving algebraicity once the companion construction theorems are granted. The core ingredient is a *realization/microstructure* scheme producing integral cycles whose calibration defect tends to 0 and whose masses converge to the cohomological lower bound. Any calibrated limit is a positive combination of complex subvarieties [11, 10], hence algebraic on projective manifolds [5, 12]. A signed decomposition reduces the general case to the cone-positive case, and Hard Lefschetz [13, 8] treats the full p-range.

Problem

Let X be a smooth projective complex variety of complex dimension n, equipped with a Kähler form ω in the Chern class of a fixed ample line bundle $L \to X$. Fix an integer $1 \le p \le n$ and a rational Hodge class

$$\gamma \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X).$$

The Hodge conjecture asks whether γ is represented by an algebraic cycle of codimension p.

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Route

Set $\psi := \omega^{n-p}/(n-p)!$, the Kähler calibration of complex $(n-p)$-planes, and $k := 2n - 2p$. For an integral k-cycle T, the *calibration defect* is

$$\mathrm{Def}_{\mathrm{cal}}(T) := \mathrm{Mass}(T) - \int_T \psi \ \geq \ 0.$$

The proof has three stages.

1. Signed decomposition. Every rational Hodge class γ decomposes as $\gamma = \gamma^+ - \gamma^-$ where γ^+ is cone-positive and $\gamma^- = N[\omega^p]$ is algebraic (Lemma 5.2).

2. Cone-positive realization (SYR). For a cone-positive class γ^+ with smooth closed cone-valued representative β, the companion papers construct integral k-cycles T_j with $\partial T_j = 0$, $[T_j] = \mathrm{PD}(m[\gamma^+])$, and $\mathrm{Def}_{\mathrm{cal}}(T_j) \to 0$ (Theorem 4.1).

3. Calibrated limit and algebraicity. A subsequential flat limit of $\{T_j\}$ is ψ-calibrated, hence a holomorphic chain by Harvey–Lawson/King, and algebraic by Chow/GAGA (Theorem 3.6).

Organization

Section 2 states the interface theorems from the companion papers (without proof). Section 3 proves the closure chain: almost-calibrated sequences have calibrated limits; SYR-realizable forms yield algebraic cycles. Section 4 states the Automatic SYR theorem (proved by combining the companion-paper interfaces). Section 5 proves the signed decomposition. Section 6 states and proves the main theorem.

Series convention. All results in Sections 3–6 are proved from the interface propositions in Section 2. Those interface results are established in the companion preprints "Bergman-scale holomorphic complete intersections with prescribed tangent directions and template-controlled face traces" (Paper I, Theorem 1.1) [1], "Coherent discretization of strongly positive forms by vertex-prefix face balancing" (Paper II, Theorem 1.1) [2], and "Fixed-class integral cycles modulo torsion from coherent calibrated packets via period-first rounding" (Paper III, Theorem 1.1) [3]. The dependence on those inputs is intended to be explicit rather than implicit.

2 Interface theorems from companion papers

The following three propositions are the imported outputs of the companion series. They are used in this paper only as explicit interface inputs.

Proposition 2.1 (H1 interface: local holomorphic manufacturing)**.** *For each mesh cell Q and each direction family prescribed by the local Carathéodory data of β on Q, there exists a ψ-calibrated sheet-sum integral current S_Q satisfying* $\mathrm{Mass}(S_Q) = \langle S_Q, \psi \rangle$ *with quantitative slope and budget control, together with template-controlled face traces in flat norm.*

Source in the series. This is exactly Theorem 1.1 of Paper I, "Bergman-scale holomorphic complete intersections with prescribed tangent directions and template-controlled face traces" [1]. □

Proposition 2.2 (H2 interface: assembly packet existence)**.** *For every sufficiently small mesh size $h > 0$, there exists an assembly packet at scale h for β satisfying the three properties needed downstream:*

(A1) *cube-wise mass and barycenter control,*

(A2) *global coherence across labels, and*

(A3) *a marginal period bound.*

Source in the series. This is exactly Theorem 1.1 of Paper II, "Coherent discretization of strongly positive forms by vertex-prefix face balancing" [2]. □

Proposition 2.3 (H3 interface: exact-class gluing and almost-calibration)**.** *Assume an assembly packet satisfying* (A1)–(A3) *as in Proposition 2.2. Then, for all sufficiently small mesh sizes, one can choose the final marginal activations so as to obtain a corrected integral current T_h with:*

(i) $\partial T_h = 0$,

(ii) $[T_h] = \mathrm{PD}(m[\gamma])$ *in* $H_k(X, \mathbb{Z})/\mathrm{tors}$,

(iii) $0 \le \mathrm{Def}_{\mathrm{cal}}(T_h) \to 0$, *and hence*

(iv) $\mathrm{Mass}(T_h) \to c_0$.

Source in the series. This is exactly Theorem 1.1 of Paper III, "Fixed-class integral cycles modulo torsion from coherent calibrated packets via period-first rounding" [3]. □

3 Closure chain

Spine theorem

Theorem 3.1 (Quantitative almost-mass-minimizing cycles)**.** *Let (X, ω) be a smooth projective Kähler manifold of complex dimension n, fix $1 \le p \le n$, and set $\psi = \omega^{n-p}/(n-p)!$ and $k := 2n-2p$. Let $[\gamma] \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$ admit a smooth closed cone-valued representative β. Choose $m \ge 1$ so that $m[\gamma] \in H^{2p}(X, \mathbb{Z})$, and set*

$$A := \mathrm{PD}(m[\gamma]) \in H_k(X, \mathbb{Z}), \qquad c_0 := \langle A, [\psi] \rangle = m \int_X \beta \wedge \psi.$$

For each mesh scale $h_j \to 0$, applying Propositions 2.1, 2.2, and 2.3 produces integral k-cycles T_j with $\partial T_j = 0$, $[T_j] = A$, and

$$0 \;\le\; \mathrm{Mass}(T_j) - c_0 \;\le\; 2\,\mathrm{Mass}(G_j),$$

where G_j is the gluing correction with $\mathrm{Mass}(G_j) \to 0$. *In particular* $\mathrm{Mass}(T_j) \to c_0$, *i.e.* $\mathrm{Def}_{\mathrm{cal}}(T_j) \to 0$.

Proof. Since ψ is closed and $[T_j] = A$, the pairing $\langle T_j, \psi \rangle = \langle A, [\psi] \rangle = c_0$ is topological. The calibration inequality $\mathrm{Mass}(T_j) \ge c_0$ gives the lower bound. Writing $S_j = T_j + G_j$ where S_j is ψ-calibrated (so $\mathrm{Mass}(S_j) = \langle S_j, \psi \rangle$), the triangle inequality yields

$$\mathrm{Mass}(T_j) \le \mathrm{Mass}(S_j) + \mathrm{Mass}(G_j) = \langle T_j + G_j, \psi \rangle + \mathrm{Mass}(G_j) \le c_0 + 2\,\mathrm{Mass}(G_j). \qquad \square$$

Microstructure/gluing estimate

Proposition 3.2 (Microstructure/gluing estimate)**.** *Let T^{raw} be the integral k-current built from the microstructure pieces on a mesh of size h, $R := \partial T^{\mathrm{raw}}$, and $\delta := \mathcal{F}(R)$. Then there exists an integral k-current R_{glue} with*

$$\partial R_{\mathrm{glue}} = -R, \qquad \mathrm{Mass}(R_{\mathrm{glue}}) \le \delta + C_X\, \delta^{\frac{k}{k-1}},$$

where C_X depends only on X and k.

Proof. By the flat-norm definition, decompose $R = R_0 + \partial Q$ with $\mathrm{Mass}(R_0) + \mathrm{Mass}(Q) \le \delta + \eta$. Then R_0 is an integral $(k-1)$-cycle bounding in X, so the Federer–Fleming filling inequality [6, 7] gives Q_0 with $\partial Q_0 = R_0$ and $\mathrm{Mass}(Q_0) \le C_X\, \mathrm{Mass}(R_0)^{k/(k-1)}$. Setting $R_{\mathrm{glue}} := -(Q + Q_0)$ and letting $\eta \to 0$ yields the bound. □

Cohomology matching

Proposition 3.3 (Integral cohomology constraints)**.** *Fix smooth closed k-forms $\Theta_1, \dots, \Theta_b$ representing an integral basis of $H^k(X, \mathbb{Z})/\mathrm{tors}$, and write*

$$I_\ell := \int_X \beta \wedge \Theta_\ell.$$

Let S be the raw integral current obtained after integer rounding, and let U_ε be an integral current with $\partial U_\varepsilon = \partial S$ and $\mathrm{Mass}(U_\varepsilon) \to 0$. Set $T_\varepsilon := S - U_\varepsilon$. By refining the mesh and choosing the integer sheet counts $N_{Q,j}$ via fixed-dimensional discrepancy rounding [4], one achieves

$$\int_{T_\varepsilon} \Theta_\ell = m\, I_\ell \qquad (\ell = 1, \dots, b),$$

so that $[T_\varepsilon] = \mathrm{PD}(m[\gamma])$ in $H_k(X, \mathbb{Z})/\mathrm{tors}$.

Proof. Cube-wise mass and barycenter control force the fractional bookkeeping current to approximate the target period vector within $\frac{1}{8}$. Discrepancy rounding adds at most $\frac{1}{8}$ more. The gluing correction contributes $< \frac{1}{4}$ to each period. Since closed integral cycles pair integrally with integral classes, the total error < 1 forces exact equality. □

Almost-calibration

Proposition 3.4 (Almost-calibration)**.** *Let S be a ψ-calibrated sheet sum and U_ε an integral current with $\partial U_\varepsilon = \partial S$ and $\mathrm{Mass}(U_\varepsilon) \to 0$. Set $T_\varepsilon := S - U_\varepsilon$. Then:*

(i) $\displaystyle\int_{T_\varepsilon} \psi = \langle \mathrm{PD}(m[\gamma]), [\psi] \rangle = c_0$ *for all ε.*

(ii) $0 \le \mathrm{Def}_{\mathrm{cal}}(T_\varepsilon) \le 2\, \mathrm{Mass}(U_\varepsilon) \to 0$.

(iii) $\mathrm{Mass}(T_\varepsilon) \to c_0$.

Proof. Item (i): ψ is closed and $[T_\varepsilon]$ is fixed, so the pairing is topological. For (ii), $\mathrm{Mass}(S) = \int_S \psi$ and the comass bound give

$$\mathrm{Def}_{\mathrm{cal}}(T_\varepsilon) = \mathrm{Mass}(T_\varepsilon) - \int_{T_\varepsilon} \psi \le (\mathrm{Mass}(S) + \mathrm{Mass}(U_\varepsilon)) - (\int_S \psi - |\int_{U_\varepsilon} \psi|) \le 2\, \mathrm{Mass}(U_\varepsilon).$$

Item (iii) combines (i) and (ii). □

Calibrated limits are holomorphic

Definition 3.5 (Stationary Young-measure realizability (SYR))**.** A cone-valued smooth closed (p,p)-form β representing $[\gamma]$ is *SYR-realizable* if there exist integral $(2n-2p)$-cycles T_k with $\partial T_k = 0$, $[T_k] = \mathrm{PD}(m[\gamma])$ for a fixed $m \geq 1$, and $\mathrm{Def}_{\mathrm{cal}}(T_k) \to 0$.

Theorem 3.6 (Realization from almost-calibrated sequences)**.** *Assume β is SYR-realizable. Then there exists an integral $(2n-2p)$-cycle T with $\partial T = 0$, $[T] = \mathrm{PD}(m[\gamma])$, and $\mathrm{Mass}(T) = \langle T, \psi \rangle = c_0$.*

In particular, T is ψ-calibrated. By Harvey–Lawson [10], T is a positive d-closed integral current of bidimension (p,p). By King's theorem [11], T is a holomorphic chain $T = \sum_j m_j [V_j]$ with V_j irreducible analytic subvarieties. If X is projective, each V_j is algebraic by Chow/GAGA [5, 12].

Proof. Since $[T_k] = \mathrm{PD}(m[\gamma])$ and ψ is closed, $\langle T_k, \psi \rangle = c_0$ for all k. The defect hypothesis gives $\mathrm{Mass}(T_k) \to c_0$, so $\sup_k \mathrm{Mass}(T_k) < \infty$. By Federer–Fleming compactness [7], passing to a subsequence, $T_k \to T$ in flat norm with T an integral cycle and $[T] = \mathrm{PD}(m[\gamma])$. Flat convergence gives $\langle T, \psi \rangle = c_0$; lower semicontinuity gives $\mathrm{Mass}(T) \leq c_0$; the comass inequality gives $\langle T, \psi \rangle \leq \mathrm{Mass}(T)$. Combining: $\mathrm{Mass}(T) = c_0 = \langle T, \psi \rangle$, so T is ψ-calibrated. The holomorphic-chain conclusion and algebraicity follow from Harvey–Lawson [10], King [11], and Chow/GAGA [5, 12, 9, 8]. □

Theorem 3.7 (Calibrated realization under SYR)**.** *Assume β is SYR-realizable. Then the conclusion of Theorem 3.6 holds: there exists a ψ-calibrated holomorphic chain $T = \sum_j m_j [V_j]$ representing $\mathrm{PD}(m[\gamma])$, and $[\gamma]$ is algebraic if X is projective.*

Proof. Apply Theorem 3.6 to the SYR sequence. □

4 Automatic SYR

Theorem 4.1 (Automatic SYR for cone-valued forms)**.** *Assume Propositions 2.1, 2.2, and 2.3. Let (X, ω) be a smooth complex projective manifold of complex dimension n, and let $1 \leq p \leq n/2$. (For $p > n/2$, reduce to the complementary degree $n - p$ by Hard Lefschetz [13, 8].) Let β be a smooth closed cone-valued (p,p)-form representing $[\gamma] \in H^{p,p}(X; \mathbb{Q})$.*

Then β is SYR-realizable: there exist integral $(2n-2p)$-cycles T_k with $\partial T_k = 0$, $[T_k] = \mathrm{PD}(m[\gamma])$, and $\mathrm{Def}_{\mathrm{cal}}(T_k) \to 0$. In particular, by Theorem 3.7, $[\gamma]$ is algebraic.

Proof. Fix a mesh parameter $\varepsilon > 0$. By Proposition 2.2, for each sufficiently small scale $\varepsilon > 0$ there exists an assembly packet satisfying (A1)–(A3). Applying Proposition 2.3 to that packet yields closed integral cycles T_ε with $[T_\varepsilon] = \mathrm{PD}(m[\gamma])$ and $\mathrm{Def}_{\mathrm{cal}}(T_\varepsilon) \to 0$ as $\varepsilon \to 0$. Choosing $\varepsilon_k \downarrow 0$ gives the SYR sequence. Theorem 3.7 concludes algebraicity. □

5 Signed decomposition

Lemma 5.1 (Strict interior positivity of the Kähler power)**.** *The (p,p)-form ω^p is strictly positive in the calibrated cone: $\omega^p(x) \in \operatorname{int} K_p(x)$ for all $x \in X$. There exists a uniform radius $r_0 > 0$ such that $B(\omega^p(x), r_0) \subset K_p(x)$ for every x.*

Proof. In a unitary frame at x, ω_x^p is a strictly (strongly) positive (p,p)-form. Compactness of X and continuity of the cone give uniform $r_0 > 0$. □

Lemma 5.2 (Signed decomposition)**.** *Let $\gamma \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ be any rational Hodge class. Then there exist cone-positive rational Hodge classes γ^+ and γ^- with $\gamma = \gamma^+ - \gamma^-$, where $\gamma^- = N[\omega^p]$ for some $N \in \mathbb{Q}_{>0}$.*

Proof. Let α represent γ, set $M := \sup_X \|\alpha\|$, and choose $N \in \mathbb{Q}_{>0}$ with $N > M/r_0$. Then $\alpha(x) + N\omega^p(x) \in K_p(x)$ for all x, so $\gamma^+ := \gamma + N[\omega^p]$ is cone-positive and $\gamma^- := N[\omega^p]$ is algebraic. □

Lemma 5.3 (ω^p is algebraic)**.** *The class $[\omega^p] = c_1(L)^p$ lies in the $\mathbb{Q}$-span of algebraic cycle classes. For $m_{\mathrm{hol}} \gg 0$, a smooth complete intersection $Z = D_1 \cap \cdots \cap D_p$ with $D_i \in |L^{\otimes m_{\mathrm{hol}}}|$ satisfies $\mathrm{PD}([Z]) = m_{\mathrm{hol}}^p[\omega^p]$.*

Proof. Very ampleness of $L^{\otimes m_{\mathrm{hol}}}$ and Bertini's theorem [9] give a smooth Z with the stated class. □

6 Main theorem

Theorem 6.1 (Cone-positive classes are algebraic)**.** *Assume Propositions 2.1, 2.2, and 2.3. Let $\gamma^+ \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ be a cone-positive rational Hodge class on a smooth complex projective manifold, with $p \le n/2$. Then γ^+ is algebraic.*

Proof. By Theorem 4.1, γ^+ is SYR-realizable. By Theorem 3.7, there exists a ψ-calibrated holomorphic chain representing $\mathrm{PD}(m[\gamma^+])$. Since X is projective, Chow/GAGA gives algebraicity. □

Conjecture 6.2 (Hodge Conjecture)**.** *Let X be a smooth complex projective manifold. Every rational cohomology class $\gamma \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ is represented by an algebraic cycle.*

Theorem 6.3 (Hodge Conjecture for rational (p,p) classes)**.** *Assume Propositions 2.1, 2.2, and 2.3. Let X be a smooth complex projective manifold. Then every rational Hodge class $\gamma \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ is algebraic.*

Proof. By Hard Lefschetz [13, 8] it suffices to treat $p \le n/2$. Apply Lemma 5.2: $\gamma = \gamma^+ - \gamma^-$ with $\gamma^- = N[\omega^p]$ algebraic (Lemma 5.3) and γ^+ cone-positive. By Theorem 6.1, γ^+ is algebraic. Since algebraic classes form a $\mathbb{Q}$-subspace, $\gamma = \gamma^+ - \gamma^-$ is algebraic. □

Corollary 6.4 (Full Hodge conjecture)**.** *Every rational (p,p) class on a smooth complex projective manifold is represented by an algebraic cycle.*

Proof. Immediate from Theorem 6.3. □

Remark 6.5 (Logical structure). Theorem 6.3 combines three inputs: (i) signed decomposition (Lemma 5.2); (ii) SYR realization of cone-positive classes (Theorem 4.1, proved using the companion-paper interfaces); (iii) the calibrated-limit closure principle (Theorem 3.6) with algebraicity via Chow/GAGA.

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